

HERM: HIERARCHICAL EVENT-ROUTED MEMORY FOR CONTEXT-CONDITIONED INTERACTIVE AVATAR GENERATION

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ABSTRACT

Interactive streaming avatar generation requires natural and continuous responses to a conversational partner. However, current chunk-wise generators rely mainly on current audio and may produce generic or repetitive motions with limited use of earlier interaction. Keeping all previous chunks is costly, while compression alone neither identifies which behaviors matter now nor ensures that retained history affects generation. We view a streaming conversation as a sequence of multimodal interaction events, in which an appropriate response depends on both the current partner signal and relevant past events. Based on this view, we propose Hierarchical Event-Routed Memory (HERM) for history-conditioned interactive avatar generation. HERM continuously organizes completed interaction events into a bounded memory and retrieves relevant trajectories according to the current audio, speaking/listening state, and recent motion. Experiments examine whether retrieved trajectories affect the next chunk, improve contextual fit, reduce repetitive behavior, and retain the avatar’s expressive traits.

Index Terms— Streaming avatar generation, multimodal interaction, memory-augmented generation.

1. INTRODUCTION

Interactive streaming avatar generation aims to produce natural, continuous, and responsive animation during conversation. Recent work enables real-time and long-form generation through streaming or autoregressive models [1, 2, 3], while existing interactive systems explicitly model participants’ speaking and listening states using dyadic audio, motion, and conversational context [4, 5, 6, 7, 8]. Yet immediate cues do not specify a unique visual response: the same partner behavior may admit several plausible reactions [9, 10, 11]. Without further context, the generator may repeatedly choose generic motions from this broad response space. Interaction history can reduce this ambiguity by favoring responses that fit both the current situation and the avatar’s established expressive patterns. This motivates history-conditioned generation, where accumulated interaction guides each new full-body response.

We view interaction history as a multimodal event stream, where each response may draw on a different subset of past events. Interactive models already use multimodal context to infer conversational states [6], while query-aware memory methods retrieve different evidence for different queries [12, 13]. The central question is therefore which past events should impact the next response, rather than simply how much history is retained.

Selective history use must remain efficient and reliable as the stream grows. Retaining the full history increases memory and computation with duration [14, 15]. Meanwhile, the history increasingly contains the model’s own outputs, so unfiltered long-term updates may reinforce repetitive or unreliable behavior. This risk is related

to exposure bias and error accumulation in autoregressive video generation [16, 17]. The memory must therefore remain bounded while controlling how generated events become long-term evidence.

To meet these requirements, we propose Hierarchical Event-Routed Memory (HERM) for history-conditioned full-body streaming avatar animation.¹ Unlike memory designed for general video coherence or 3D head motion style [12, 18], HERM represents completed multimodal interaction as behavioral events for guiding the next response. It organizes the initial demonstration and completed chunks into a bounded event hierarchy, then routes relevant trajectories using current audio, speaking/listening states, and recent motion. Each completed chunk updates recent memory, while calibrated admission controls long-term writes from generated content. This conditioned-read, controlled-write design allows history to evolve without unbounded cost or unchecked feedback.

Our contributions are summarized as follows:

- **Event-Structured Memory.** We organize multimodal history into a bounded event hierarchy, retaining temporal structure beyond local context.
- **Conditioned Read, Controlled Write.** We condition retrieval on current cues and calibrate long-term admission of generated events through paired rollouts.
- **Controlled Validation.** Our approach improves contextual appropriateness and reduces repetition over the baseline, with ablations revealing content-dependent responses.

2. METHOD

HERM adds bounded event memory to our trained streaming interactive avatar, which takes a reference image and streaming audio as input. A demonstration video initializes memory before generation.

2.1. From Chunks to Event Memory

For each chunk, HERM reads history under current conditions, generates the chunk, and updates memory upon completion:

$$\begin{aligned} M_k &= \mathcal{R}(\mathcal{H}_{k-1}, c_k), \hat{X}_k = G(C_k; M_k), \\ \mathcal{H}_k &= \mathcal{U}(\mathcal{H}_{k-1}, \hat{X}_k, C_k). \end{aligned} \quad (1)$$

Here, $\mathcal{H}_{-1}$ is the initialized memory, C_k contains the available conditions for chunk k , and c_k summarizes them for retrieval. The backbone G remains frozen, while trainable memory readers incorporate the bounded readout M_k .

To update memory, a memory writer extracts aligned visual, audio, and state features from completed chunks, omitting frames repeated across adjacent chunks. A weighted boundary score com-

¹Project page: <https://github.com/Wx-Fu/HERM>

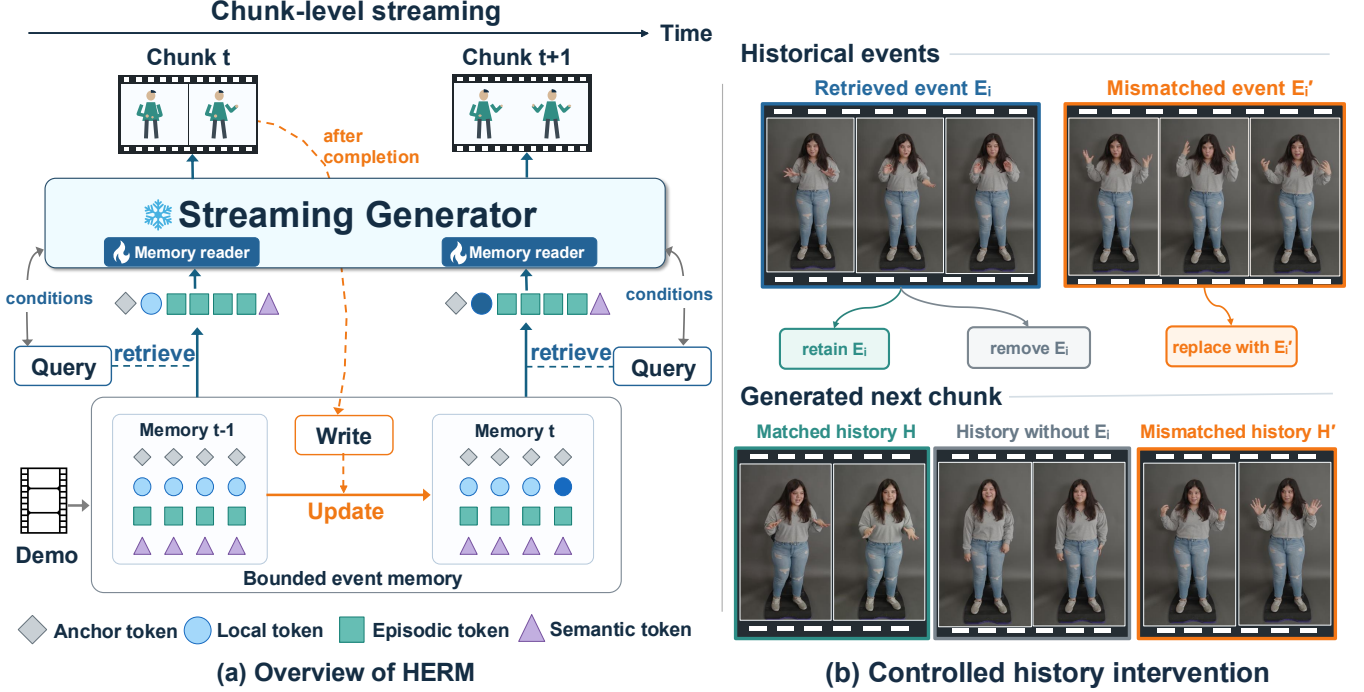


Fig. 1. Bounded event memory and controlled history intervention in HERM. (a) A demonstration initializes memory; current conditions retrieve a fixed-size readout before each chunk, which updates memory after completion. (b) We retain, remove, or replace event E_i with E'_i while holding other history and generation conditions fixed, then compare the next chunks.

bins novelty against the open event, audio changes, and state transitions; exceeding a single threshold starts a new event, otherwise the chunk extends the open event.

Each event stores four value fields: onset, core, offset, and prototype, which summarize event dynamics rather than a full motion sequence. For each chunk, the memory writer forms local tokens by combining pooled visual latents with self/partner audio, state, and temporal/spatial positions. Four learned queries with temporal masks extract onset, core, offset, and prototype from the first, middle, last, and full valid windows, respectively. Latent frames shared by adjacent chunks are deduplicated during writing, so each overlapping interval contributes only once. For an event spanning multiple chunks, onset is taken from its first valid chunk and offset from its latest; core and prototype accumulate the intervening valid content through learned-gate, duration-weighted updates.

HERM maintains three capacity-bounded banks: Local stores recent features, Episodic event summaries, and Semantic selected older events. An immutable Anchor bank preserves reference appearance and scene information. Eviction protects recent events and minimizes a retention score rewarding importance, reliability, usage, and transition coverage while penalizing redundancy and age.

2.2. Conditioned Read, Controlled Write

As shown in Fig. 1(a), current conditions feed both the streaming generator and Query path. The Query summarizes self and partner audio, states, and the generator’s motion prefix to form c_k . It conditions Local and Semantic resampling and ranks Episodic events. HERM retains $K \leq 6$ Episodic events, each represented by four tokens, under a total readout budget B :

$$B = B_A + B_L + 4K + B_S. \quad (2)$$

The default setting uses $B = 64$ and $K = 6$. These event tokens, the resampled tokens, and Anchor tokens form the bounded readout, which enters the generator through gated cross-attention.

At read time, event tokens receive encodings of relative time, duration, and field role. Readers use importance, reliability, transition type, and observed/generated source to interpret events. Relative time is recomputed from stored timestamps, leaving persistent history independent of the current query.

Training uses hard top-six selection in the forward pass and a differentiable assignment over all valid candidates for straight-through gradients [19]. Inference uses hard selection alone. Missing event slots are masked.

After each chunk is finalized, its non-overlapping content updates Local and the current Episodic event. When a new event would exceed Episodic capacity, the retention policy evicts an existing event. An evicted observed event from the demonstration history is consolidated into Semantic through gated merging, whereas an evicted generated event enters Semantic only if it passes calibrated admission and is otherwise discarded. Anchor remains immutable, while recent generated chunks remain available in Local and the active Episodic event until eviction.

2.3. Memory Distillation and Closed-Loop Adaptation

Warm-up trains the memory writer, consolidator, full reader, and confidence head; the head maps motion jump, reference drift, and writer event summaries to a quality logit under Smooth-L1 and ranking supervision from suffix-normalized latent errors. After warm-up, an exponential-moving-average full reader supervises the bounded reader. The teacher bypasses readout compression but shares storage and eviction constraints. The bounded path starts with a learned generic query before conditioning on c_k ; adaptation

Table 1. Main comparison on Seamless Interaction. ↓ and ↑ indicate lower and higher is better, respectively. Results are compared only within the same output scope. The best result in each scope is shown in **bold** with a light-blue background.

Method	Scope	Objective metrics						Human ratings		
		FID↓	FVD↓	CSIM↑	Sync-C↑	rPCC (%)↓	cMRS↓	Ctx.↑	Nat.↑	ID Cons.↑
<i>Interactive talking-head generation</i>										
Avatar Forcing [20]	Head	374.553	303.523	0.747	1.893	12.099	1.721	3.225	3.125	3.100
DyStream [21]	Head	380.115	323.337	0.817	2.670	10.189	1.257	3.056	2.389	2.528
<i>Body-visible audio-driven animation</i>										
OmniAvatar [22]	Body	90.056	94.617	0.691	3.220	10.792	1.259	3.633	4.100	4.088
LiveAvatar [23]	Body	86.879	99.469	0.651	3.191	9.579	1.102	3.062	2.500	2.812
StableAvatar [24]	Body	83.393	94.711	0.495	1.237	9.669	1.129	3.167	2.792	2.538
G_0 (base generator)	Body	86.684	83.159	0.878	3.694	9.372	1.074	3.577	4.091	4.227
HERM (Ours)	Body	85.497	80.925	0.873	3.767	7.239	0.836	4.023	4.114	4.114

updates the router, resamplers, and readers while freezing the writer, consolidator, and backbone. The adaptation objective is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{roll}}\mathcal{L}_{\text{roll}} + \lambda_{\text{dist}}\mathcal{L}_{\text{distill}} + \lambda_{\text{rel}}\mathcal{L}_{\text{rel}}. \quad (3)$$

Here, $\mathcal{L}_{\text{FM}}$ is the velocity loss. $\mathcal{L}_{\text{roll}}$ matches low-frequency motion changes across rollout boundaries. $\mathcal{L}_{\text{distill}}$ matches bounded-reader velocity predictions with the full reader’s on the same noisy inputs, excluding the fixed prefix. $\mathcal{L}_{\text{rel}}$ aligns router probabilities with detached teacher event attention.

Late training uses autoregressive closed-loop rollouts: each generated chunk updates Local and Episodic and, under a frozen provisional calibrator, may update mutable Semantic slots before the next chunk. We mix blocked, calibrated, and deliberately over-admitted writes to expose the reader to deployment-like history.

After training, we freeze the model and recalibrate admission using multi-step paired rollouts:

$$H_i^{(L)} = \mathcal{M}_{1:L}(\text{admit } i) - \mathcal{M}_{1:L}(\text{block } i). \quad (4)$$

The paired branches share pre-write memory, conditions, and noise and differ only in initial admission; both follow the same policy for L chunks. The lower-is-better score $\mathcal{M}$ is a fixed weighted sum of normalized rollout inconsistency, identity drift, boundary discontinuity, and state-response error. An admission is harmful when this difference exceeds h_{tol} .

Using disjoint held-out splits, we fit Platt scaling from the frozen head’s quality logits to the resulting safety labels and select threshold τ . After closed-loop validation, inference admits events whose calibrated safety probability exceeds τ .

3. EXPERIMENTS

3.1. Experimental Setup

Dataset. We train and evaluate on Seamless Interaction [25] with participant-disjoint splits, where one participant is the target avatar and the other the interaction partner. HERM initializes its memory from a non-overlapping target clip.

Baselines and protocol. We compare against interactive talking-head methods Avatar Forcing [20] and DyStream [21], and the body-visible audio-driven methods OmniAvatar [22], LiveAvatar [23], and StableAvatar [24]. We use a dyadic interactive talking-video model built on the Wan 14B video diffusion Transformer (DiT) as the generation backbone; G_0 denotes this model without HERM memory. We follow official inference settings and align the test identity, audio

interval, frame rate, and resolution whenever the interfaces permit. Table 1 separates head-only and body-visible outputs; scores and boldface are compared only within the same scope.

Metrics. We report FID [26] and FVD [27] (lower is better) for frame and video quality, CSIM from ArcFace embeddings [28] for identity preservation, and Sync-C [29] for lip synchronization. To measure whether the generated response preserves partner–avatar motion coupling, we adapt rPCC [30] to whole-body motion:

$$\text{rPCC} = \frac{1}{D} \sum_{d=1}^D |\rho(v_d^p, v_d^{\text{gt}}) - \rho(v_d^p, v_d^{\text{gen}})|. \quad (5)$$

where v^p , v^{gt} , and v^{gen} are root-aligned, body-scale-normalized pose velocities of the partner, ground-truth target, and generated target. We extract whole-body keypoints with DWPose [31]. Tables report $100 \times \text{rPCC}$ as a percentage; head-only methods use shared facial and head-pose channels and are compared only within scope.

We measure long-range repetition with a GT-calibrated Motion Repetition Score:

$$\text{cMRS} = R + \frac{1}{2} \left| \log \frac{E_g + \epsilon}{E_t + \epsilon} \right| + \frac{1}{2} [C_t - C_g]_+, \quad (6)$$

where R is the mean maximum cosine similarity between normalized 28-frame pose–velocity chunks at least 3.36 s apart; E and C denote motion energy and active-chunk coverage for generated (g) and ground-truth (t) motion. State-specific coverage thresholds are fixed on validation data.

Human Study. Twenty raters scored 12 anonymized clips (12 s–2 min) on a five-point scale for contextual appropriateness/partner responsiveness, naturalness/continuity, and identity consistency. The synchronized partner context was shown for contextual assessment, while a reference image or initialization clip was provided for identity assessment.

3.2. Main Comparison

Table 1 compares HERM with existing streaming avatar methods. Since head-only and body-visible systems have different output scopes, we focus the main comparison on body-visible animation. HERM achieves the best FVD, Sync-C, rPCC, cMRS, contextual appropriateness, and naturalness among the body-visible methods. The largest gains concern interaction behavior: compared with the controlled backbone G_0 , rPCC decreases from 9.372% to 7.239%, cMRS from 1.074 to 0.836, and contextual appropriateness increases

Table 2. History interventions and ablations. All Δ values use the same 160 samples. Best rPCC and cMRS results are highlighted.

Variant	Δ_{read}	Δ_{shuf} (%)	Δ_{mis} (%)	rPCC (%) $\downarrow$	cMRS $\downarrow$
G_0 (no history)	N/A	N/A	N/A	9.372	1.074
Full-bank reader	2.137	1.265	3.546	11.619	1.386
Fixed query	0.126	2.683	3.934	12.226	1.020
Flat chunk memory	1.683	1.265	4.809	9.579	1.086
w/o online update	0.882	0.232	0.480	9.372	1.072
Ungated admission	1.421	1.265	0.850	11.729	1.051
HERM	1.665	5.258	4.350	7.239	0.836

from 3.577 to 4.023. These changes indicate stronger partner–avatar coupling, less repetitive motion, and better contextual fit.

The improvement is not limited to response metrics. HERM reduces FVD from 83.159 to 80.925 and slightly improves FID over G_0 , while maintaining comparable naturalness. Although its CSIM and identity-consistency rating are slightly lower than those of G_0 (0.873 vs. 0.878 and 4.114 vs. 4.227), the differences are small relative to the gains in response metrics. Overall, HERM improves partner responsiveness and motion diversity without substantial loss of visual quality.

3.3. Does HERM Use Relevant History?

We test whether retrieved history is readable and useful by fixing the inputs and noise. The Anchor-only condition removes retrieved history. The shuffled condition reassigns all four raw Episodic fields while preserving metadata, whereas the content-replaced condition substitutes same-identity events matched by role, duration, and motion energy. Figure 1(b) visualizes these interventions: retaining, removing, or replacing E_i changes the next generated chunk while remaining history and conditions are fixed.

We measure direct output influence with

$$\Delta_{\text{read}} = \left[\frac{1}{|\mathcal{V}|} \sum_{u \in \mathcal{V}} \left(\Delta \tilde{P}_{\text{true}}(u) - \Delta \tilde{P}_{\text{anchor}}(u) \right)^2 \right]^{1/2}, \quad (7)$$

where u indexes valid time, joint, and coordinate entries of the first-differenced, hip-root-aligned, shoulder–hip-scale-normalized DW-Pose coordinates. A larger Δ_{read} indicates stronger history influence, but not necessarily a better response.

We measure content dependence using

$$\begin{aligned} \Delta_{\text{shuf}} &= \text{rPCC}_{\text{shuffled}} - \text{rPCC}_{\text{true}}, \\ \Delta_{\text{mis}} &= \text{rPCC}_{\text{replaced}} - \text{rPCC}_{\text{true}}. \end{aligned} \quad (8)$$

Here, “true” denotes intact-history generation. Since lower rPCC is better, positive gaps indicate that the intact history yields a better response than the corresponding intervention. All interventions use the same 160 valid samples. The rPCC column reports $100 \times \text{rPCC}$, while the two gaps use unscaled paired rPCC values.

Table 2 separates history influence, content dependence, and usefulness. Simply exposing the generator to more history is insufficient: the full-bank reader changes output most but degrades rPCC and cMRS, indicating unselective history use. HERM instead depends on the retrieved event content: shuffling or replacing history raises rPCC from 7.239% to 12.497% and 11.589%, respectively. Flat memory has a comparable replacement gap but worse quality, so content dependence alone is insufficient. Fixed-query retrieval also fails to adapt memory to the current condition. Removing online updates nearly eliminates the intervention gaps, while ungated

Table 3. Readout-budget and long-rollout results. The shaded row is the default; Δ Lat. is relative to G_0 .

Setting	FVD $\downarrow$	rPCC (%) $\downarrow$	cMRS $\downarrow$	Δ Lat. (ms) $\downarrow$
<i>Readout budget (tokens)</i>				
16	80.067	9.590	0.901	241
32	81.945	8.424	0.856	259
64 (default)	80.925	7.239	0.836	283
128	81.159	7.990	0.821	311
<i>Long rollout (64-token readout)</i>				
Early	79.939	7.320	0.828	283
Middle	80.990	7.186	0.831	283
Late	82.041	7.211	0.846	286

admission degrades response quality. Together, these results support event structure, condition-guided retrieval, and controlled online updates for responsive, less repetitive generation.

3.4. Efficiency and Long-Stream Stability

Table 3 identifies 64 tokens as a practical operating point. Increasing the readout from 16 to 64 tokens reduces rPCC from 9.590 to 7.239% and cMRS from 0.901 to 0.836, with 42 ms additional relative latency and only a 1.1% increase in FVD. Expanding the budget to 128 tokens improves cMRS by only another 1.8%, while rPCC worsens by 10.4% and latency increases by 28 ms. Thus, most of the benefit is obtained by 64 tokens; exposing the generator to more history does not consistently improve the response and may dilute relevant trajectories with redundant context.

For the 64-token setting, we evaluate five-minute test videos whose rollouts reach memory saturation and undergo repeated event eviction and merging. From Early to Late, relative latency remains nearly constant at 283–286 ms, while rPCC stays within 7.186–7.320% and cMRS within 0.828–0.846, indicating no progressive degradation in the measured interaction behavior. FVD increases from 79.939 to 82.041 (2.6%), suggesting modest visual drift. These results support bounded read cost and stable behavior over the evaluated long-rollout horizon.

4. CONCLUSION

Streaming interactive avatars can generate visually coherent, continuous animation with reasonable appearance quality, yet their responses often fall back on generic motions. The challenge is therefore not merely producing plausible movement: immediate cues leave several valid visual choices, while the absence of relevant interaction history makes it difficult to select a response suited to the ongoing exchange. We introduced HERM to address this problem by organizing completed chunks into bounded multimodal events, retrieving relevant history under current conditions, and controlling how generated events are consolidated into long-term memory. Controlled interventions and ablations show that selected event content changes subsequent chunks, while structured updates and calibrated admission make this influence useful. Experiments show improved contextual appropriateness and partner–avatar motion coupling, less repetitive motion, and comparable appearance quality, with stable read cost and behavior over the evaluated long rollouts. Overall, HERM shows that selective and controlled interaction memory can make streaming avatar animation more responsive and varied without requiring unbounded history.

5. REFERENCES

- [1] Le Shen et al., “Soulx-flashtalk: Real-time infinite streaming of audio-driven avatars via self-correcting bidirectional distillation,” *arXiv:2512.23379*, 2025.
- [2] Dingcheng Zhen et al., “Teller: Real-time streaming audio-driven portrait animation with autoregressive motion generation,” in *CVPR*, 2025, pp. 21075–21085.
- [3] Tan Yu et al., “Soulx-flashhead: Oracle-guided generation of infinite real-time streaming talking heads,” *arXiv:2602.07449*, 2026.
- [4] Ziqiao Peng et al., “Dualtalk: Dual-speaker interaction for 3d talking head conversations,” in *CVPR*, 2025, pp. 21055–21064.
- [5] Yongming Zhu et al., “Infpr: Audio-driven interactive head generation in dyadic conversations,” in *CVPR*, 2025, pp. 10667–10677.
- [6] Ying Guo, Xi Liu, Cheng Zhen, Pengfei Yan, and Xiaoming Wei, “Arig: Autoregressive interactive head generation for real-time conversations,” in *ICCV*, 2025, pp. 12956–12965.
- [7] Xuangeng Chu, Ruicong Liu, Yifei Huang, Yun Liu, Yichen Peng, and Bo Zheng, “Unils: End-to-end audio-driven avatars for unified listening and speaking,” in *CVPR*, 2026, pp. 25142–25152.
- [8] Zhiyao Sun et al., “Streamavatar: Streaming diffusion models for real-time interactive human avatars,” in *CVPR*, 2026, pp. 10887–10897.
- [9] Evonne Ng et al., “Learning to listen: Modeling non-deterministic dyadic facial motion,” in *CVPR*, 2022, pp. 20363–20373.
- [10] Cheng Luo et al., “Reactface: Online multiple appropriate facial reaction generation in dyadic interactions,” *IEEE Trans. Vis. Comput. Graph.*, vol. 31, no. 9, pp. 6190–6207, 2024.
- [11] Yinuo Wang, Yanbo Fan, Xuan Wang, Guo Yu, and Fei Wang, “Diffusion-based realistic listening head generation via hybrid motion modeling,” in *CVPR*, 2025, pp. 15885–15895.
- [12] Sihui Ji et al., “Memflow: Flowing adaptive memory for consistent and efficient long video narratives,” *arXiv:2512.14699*, 2025.
- [13] Hongyu Qu et al., “Beyond retrieval: Progressive latent memory evolution for streaming video understanding,” *arXiv:2609.04131*, 2026.
- [14] Roberto Henschel et al., “Streamingt2v: Consistent, dynamic, and extendable long video generation from text,” in *CVPR*, 2025, pp. 2568–2577.
- [15] Shuai Yang et al., “Longlive: Real-time interactive long video generation,” *arXiv:2509.22622*, 2025.
- [16] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman, “Self forcing: Bridging the train-test gap in autoregressive video diffusion,” in *NeurIPS*, 2025.
- [17] Jiaying Cui et al., “Self-forcing++: Towards minute-scale high-quality video generation,” in *ICLR*, 2026.
- [18] Yinuo Wang et al., “Mimictalker: A multimodal interactive and memory-enhanced framework for real-time dyadic 3d head generation,” in *CVPR*, 2026, pp. 32399–32409.
- [19] Yoshua Bengio, Nicholas Léonard, and Aaron Courville, “Estimating or propagating gradients through stochastic neurons for conditional computation,” *arXiv preprint arXiv:1308.3432*, 2013.
- [20] Taekyung Ki et al., “Avatar forcing: Real-time interactive head avatar generation for natural conversation,” *arXiv:2601.00664*, 2026.
- [21] Bohong Chen and Haiyang Liu, “Dystream: Streaming dyadic talking heads generation via flow matching-based autoregressive model,” *arXiv:2512.24408*, 2025.
- [22] Qijun Gan et al., “Omniavatar: Efficient audio-driven avatar video generation with adaptive body animation,” *arXiv:2506.18866*, 2025.
- [23] Yubo Huang et al., “Live avatar: Streaming real-time audio-driven avatar generation with infinite length,” *arXiv:2512.04677*, 2025.
- [24] Shuyuan Tu et al., “Stableavatar: Infinite-length audio-driven avatar video generation,” *arXiv:2508.08248*, 2025.
- [25] Vasu Agrawal et al., “Seamless interaction: Dyadic audiovisual motion modeling and large-scale dataset,” *arXiv preprint arXiv:2506.22554*, 2025.
- [26] Martin Heusel et al., “Gans trained by a two time-scale update rule converge to a local nash equilibrium,” *Advances in neural information processing systems*, vol. 30, 2017.
- [27] Thomas Unterthiner et al., “Towards accurate generative models of video: A new metric & challenges,” *arXiv preprint arXiv:1812.01717*, 2018.
- [28] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou, “Arcface: Additive angular margin loss for deep face recognition,” in *CVPR*, 2019, pp. 4690–4699.
- [29] Joon Son Chung and Andrew Zisserman, “Out of time: automated lip sync in the wild,” in *Asian conference on computer vision*, 2016, pp. 251–263.
- [30] Luchuan Song, Guojun Yin, Zhenchao Jin, Xiaoyi Dong, and Chenliang Xu, “Emotional listener portrait: Neural listener head generation with emotion,” *arXiv preprint arXiv:2310.00068*, 2023.
- [31] Zhendong Yang, Ailing Zeng, Chun Yuan, and Yu Li, “Effective whole-body pose estimation with two-stages distillation,” in *ICCVW*, 2023, pp. 4212–4222.